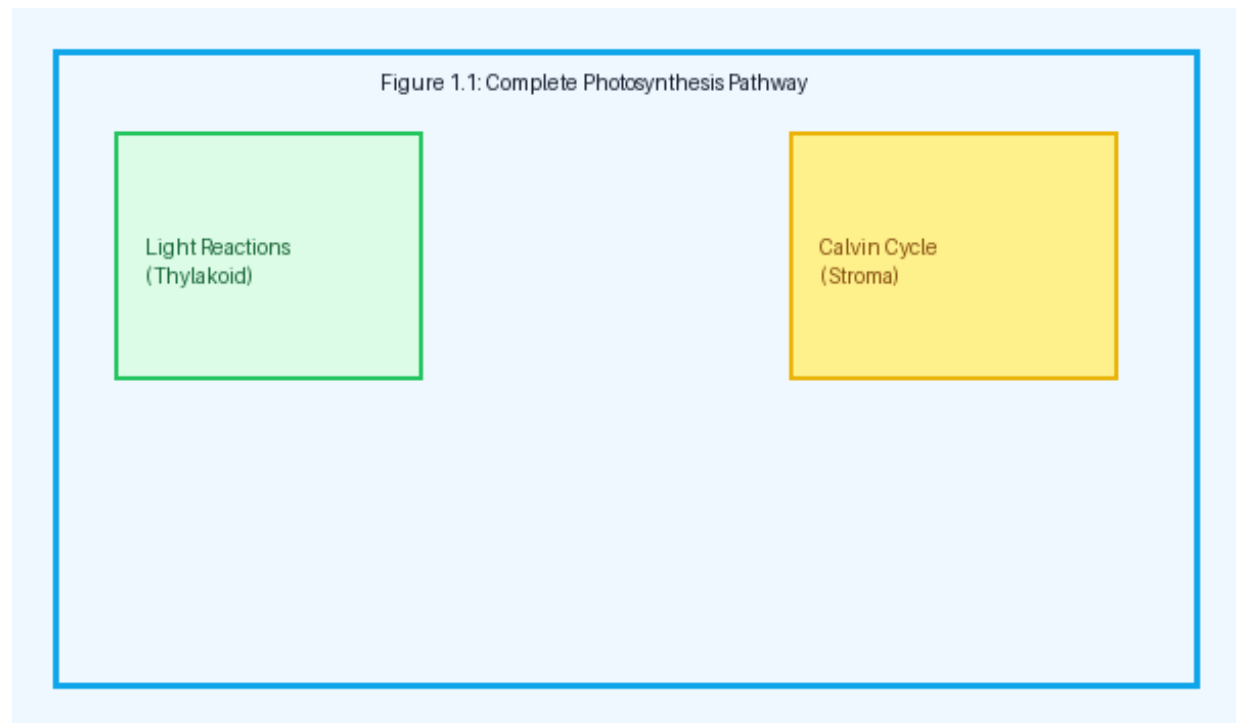


Chapter 1: Photosynthesis & Energy Conversion

Photosynthesis is the process by which green plants transform light energy into chemical energy.

Figure 1.1 illustrates the light-dependent reactions in thylakoids and the Calvin cycle in the stroma.

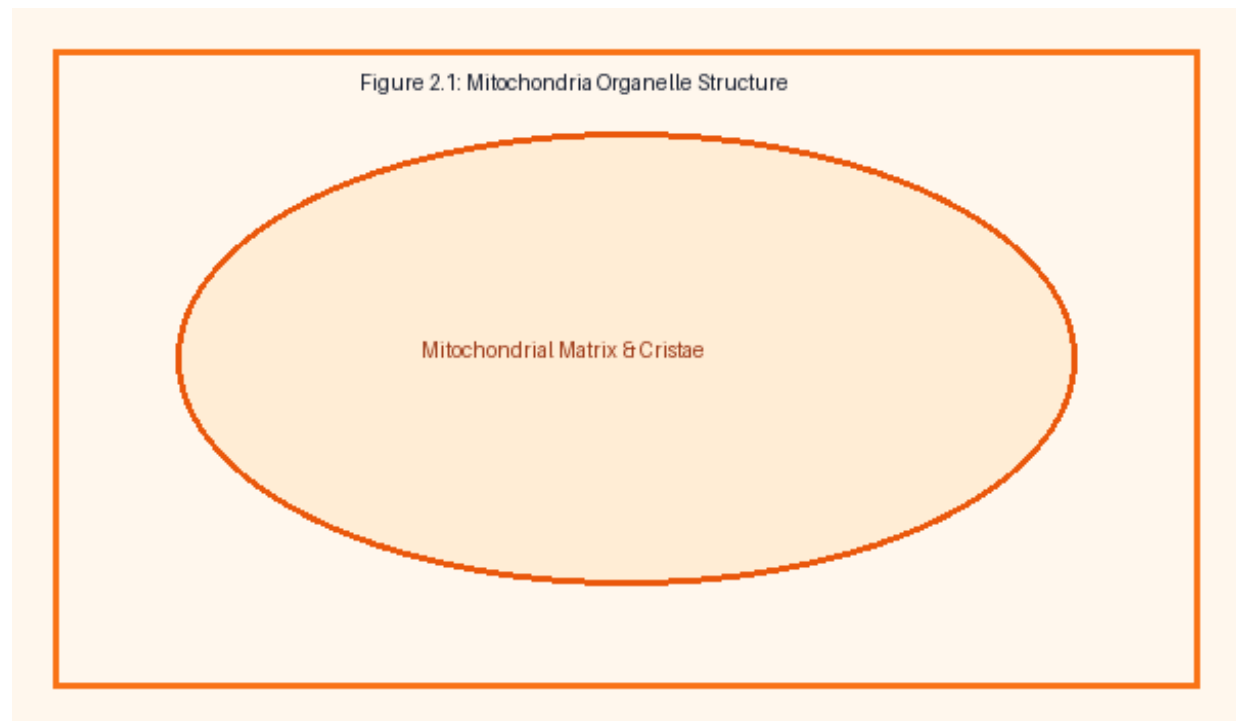


Key Equation: $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{Light} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.

Chapter 2: Cellular Respiration and Mitochondria

Mitochondria generate most of the chemical energy needed to power biochemical reactions.

Figure 2.1 shows the inner membrane folding into cristae to maximize ATP synthesis surface area.



ATP synthesis occurs along the electron transport chain located in the inner membrane.